

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester II Examination, 2023-24

Industrial Microbiology**Paper: IMB-201 — Environmental Microbiology and Biotechnology**

Time: Three Hours

Maximum Marks: 70

Instructions: Answer five questions including Question 1 which is compulsory. The figures in the right-hand margin indicate marks.

Question 1. Answer the following (not more than 50 words each): [10 × 1 = 10]

- (a) What is the recognition sequence, restriction site and the fragments produced by the restriction enzyme KpnI?
- (b) Comment on BOD and COD.
- (c) Give an example of a polyextremophile.
- (d) What are inclusion bodies?
- (e) Name a plant that degrades pesticides.
- (f) Give an example of a halophile.
- (g) What is associative mutualism? Provide an example.
- (h) What is the organic loading rate?
- (i) Name the micro-organism responsible for causing bloody diarrhoea in infants.
- (j) Name the channel involved in the regulation of glycerol content of a cell.

Question 2. What are GM crops and their types? Discuss in detail the regulation of safety of biotechnology procedures and products in India. [15]

Question 3. What are the different methods of wastewater treatment? Describe the role of micro-organisms in sludge digestion and oxidation ponds with the help of suitable diagrams. [15]

Question 4. Write short notes on any two of the following: [2 × 7.5 = 15]

- (a) Trickling filter
- (b) Types of vectors
- (c) Biofilm forming microbes

Question 5. Explain in detail the comparative stress adaptation mechanisms of acidophiles and alkaliphiles. [15]

Question 6. Write critical comments on any two of the following: [2 × 7.5 = 15]

- (a) Microbial degradation of oil spills
- (b) Sampling techniques for studying aeromicroflora of indoor environments
- (c) Recombinant protein production

Question 7. What is *in-situ* bioremediation? Briefly describe microbial degradation of synthetic pesticides with the help of suitable examples. [15]

Question 8. Write short notes on any two of the following: [2 × 7.5 = 15]

- (a) Biopesticides
- (b) Sources of indoor air pollution
- (c) Biomethanation

Question 9. Discuss in detail the different types of biosensors and the benefits of using them in Microbiology. [15]

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